



Enhancing Resume Screening Through Multi-stage LLM Classification and Hybrid Summarization

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Abstract. The increasing volume of resume applications creates a significant barrier in front of traditional resume filtering approaches, more often leading to inefficiencies, biases, and inhomogeneous candidate evaluations. In this paper, we propose a novel multi-stage architecture for resume filtering automation utilizing large language models (LLMs) in the tasks of classification and hybrid summarization to facilitate efficiency and fairness. The method integrates LLM-driven zero-shot classification, a multi-agent evaluation framework, and a hybrid summarization module combining extractive and abstractive methods. Compared with a simulated HR ground truth set, the rule-based classifier we propose demonstrated a remarkable precision level of 88.77% and a precision level of 78.39%, qualifying it as a candidate filtering instrument with a strong base. On the other hand, a more semantically complex multi-agent system meant for holistic evaluation showed a 27.92% drop in recall because of its conservative consensus mechanism but has superior modularity and interpretability characteristics critical in high-stakes decision-making scenarios. This paper contributes towards the development of automated recruiting by providing a clear and malleable architecture and placing a strong focus on performance metric vs. explainability trade-offs by utilizing artificial intelligence in recruiting applications.

Keywords: Resume Screening · LLM · Rule-based · Multi-agent · BERT

1 Introduction

The current scenario of talent hiring, organizations are being overwhelmed by increasingly higher numbers of resumes, and efficient and effective screening of candidates has become a daunting challenge. The conventional manual screening process is time-consuming, prone to errors, and most likely fails to result in fair and consistent assessments because of embedded human prejudice and time limitations. In order to contend with these inadequacies, automated resume screening

technologies emerged as a potential solution [1–3], leveraging natural language processing (NLP) [4] and machine learning (ML) [5] to scan, rank, and screen candidates against predetermined criteria.

Recent breakthroughs in Natural Language Processing (NLP), most notably concerning the onset of large language models (LLMs) like BERT and GPT, hold unprecedented potential for redefining the resume screening process [6]. These models possess an astounding understanding of linguistic subtleties and contextual features, which makes them appropriate for high-level tasks like classification, summarization, and semantic alignment. However, despite their potential, there exists relatively little work dedicated to applying LLMs in the domain of resume screening, which faces several practical and ethical challenges, including those concerning computational efficiency, limited interpretability, and potential bias in high-stakes decision-making scenarios.

Despite the latest breakthroughs that were noted in natural language processing, besides utilizing high-grade language models for various classification problems, resume evaluation remains a process that is marked by serious challenges. Such challenges can sum up in the following ways:

1. **Lack of Deep Understanding of Contextual Undertones:** Most modern resume assessment systems are incapable of much more than superficially understanding contextual factors embedded in applicants' work experiences. As a result, systems are based almost entirely on surface-level text attributes, which yields faulty or partial assessments of candidates' fit.
2. **Scalability vs Interpretability:** Modern high-performance large language models (LLMs) attain high classification accuracy levels, but they are highly computationally intensive and operate as black boxes, providing little insight into their decision-making processes.
3. **Partial Human-Like Summary:** Other systems attempt to condense resumes, but they often are just one of two extremes: completely extractive systems that just reproduce certain paragraphs of the paper or too abstractive systems that result in skipping relevant information. As a result, this diminishes the accuracy and efficiency of recruiters in assessing the suitability of applicants.
4. **Drawbacks of One-Shot Classification Methods:** One-stage classification approaches are often too stiff and fall short of developing appropriate robustness to counteract variations in resumes, candidate profiles, and fitness for various job roles.

Therefore, it's necessary to develop a filtering mechanism for resumes that utilizes a multi-level classification strategy, backed by large language models (LLMs), to filter and fine-tune candidate evaluations systematically; generate concise yet interpretable summaries via a mix of extractive and abstractive summarization methods; and achieve a tradeoff between accuracy, interpretability, and computation efficiency to guarantee practical feasibility. Resolving them will have a great impact on the quality, fairness, and transparency of automated recruitment processes, and in turn, will help organizations make wiser staffing decisions while also enhancing candidates' experiences. Our current work aims

to answer a number of key questions: RQ1: How does a multi-tiered classification architecture based on large language models (LLMs) improve both precision and contextual understanding of resume assessment compared to one-tier models? RQ2: Which type of summarization, extractive or abstractive, yields more helpful information to recruiters while ensuring semantic coherence? RQ3: How can explainability mechanisms be embedded within LLM-driven systems in a way that preserves computational efficiency? RQ4: How much are identified demographic biases by bias-detecting layers alleviated when they pass through the candidate selection process? RQ5: How does the introduced architecture fare in different industries and resume types compared to current tools?

To address these questions, our research objectives are:

- Create a full-fledged, multi-stage pipeline for a big language model that rigorously screens resumes based on their cultural fit, skill match, and eligibility criteria.
- Create a hybrid summarization module that combines large language model (LLM) abstractive methods, including GPT-4, with semantic extraction approaches, including TextRank, to construct brief but accurate candidate summaries.
- Create saliency maps concurrently with a concise attention visualization to facilitate model interpretability for recruitment specialists.
- Apply adversarial debiasing during LLM fine-tuning to reduce age, race, and gender biases.
- Determine if the model scales and if it's acceptable to apply to datasets from different domains, such as technology, healthcare, or engineering.

Aims to close the gap between high-performing deep learning structures and requirements for interpretable, balanced, and efficient resume evaluation. It enhances accuracy, reduces bias, and fosters transparency for hiring managers through advanced summarization and explanation interfaces. It sets a new benchmark for recruitment systems based on artificial intelligence.

2 Related Work

Recent studies have examined a range of approaches, frameworks, and ethical issues in the application of artificial intelligence (AI) to resume screening and hiring. The primary findings from the given literature are outlined in this section.

2.1 AI-Powered Resume Screening Frameworks and Methodologies

A multi-agent framework involving Large Language Models (LLMs) for systematic resume processing and evaluation was introduced by Lo et al. [7]. Their system includes an evaluator, score formatter, summarizer, and resume extractor. They integrated Retrieval-Augmented Generation (RAG) into the resume evaluator for enhancing contextual relevance. This enabled the incorporation of external knowledge sources like domain-specific expertise, certifications, and

company hiring criteria. By enabling dynamic adjustment of screening parameters without LLM retraining or fine-tuning, this modular design aims to enhance explainability and flexibility for diverse hiring criteria. Schnabel et al. [8] suggested a modular classification pipeline and focused on the evaluation of relevance assessment methods using LLMs. Their work benchmarked various LLM configurations (single-stage and multi-stage), e.g., GPT-4o, GPT-4o mini, Llama 3.1, and Claude 3.5 Sonnet, on the TREC Deep Learning Track (TREC-DL 23) dataset. Comparing their multi-stage pipeline, fusing GPT-4o mini and GPT-4o with solo GPT-4o, they demonstrated lower costs and better accuracy (9.7% improvement in Krippendorff's). This indicates that efficiency is boosted by fusing smaller models for binary classification prior to larger models conducting detailed evaluations.

Licari et al. [9] proposed a multi-step prompt method for LLM-based Q&A systems in bank supervisory regulation. Their process enhances the context for GPT-4o by determining applicable regulations and adding response examples. This strategy considerably enhanced regulatory Q&A precision, with a 12.5% increase in correctness and 81% recall in legal reference retrieval, demonstrating the advantages of enriched context in accurately generating responses. Pina et al. [10] examined eye-tracking data coupled with machine learning to forecast recruiter resume approval. Their pipeline, leveraging features from recruiter eye-tracking data and a Random Forest classifier, resulted in an AUC of 0.767 in predicting if a recruiter would pass a resume to the next recruitment stage. The study indicates the promise of non-traditional data sources in explaining and predicting recruiter behavior. Kaushik et al. [11] presented enhancing HR function productivity through AI. They recommended an ML model utilizing a "Greedy approach" with Kruskal's Algorithm for effective resume screening. Additionally, they recommended Knapsack algorithms for aptitude test generation and candidate weight allocation for the best fit, asserting a reduction of 75% in time-to-hire and 70% in cost per hire, and an improvement in candidate quality and diversity. Sharma and Choudhary [12] experimented with an "Elite Bag-of-Words" strategy for resume classification. By employing an iterative Maximum Entropy Partitioning (MEP) algorithm to extract "Elite keywords" from resumes and a Random Forest classifier, they attained a 62.60% accuracy, surpassing conventional TF and TF-IDF techniques.

2.2 Bias and Fairness in AI-Driven Hiring

: Bias is a major problem in AI recruiting. Massive Text Embedding (MTE) models were used in resume audit research conducted by Wilson and Caliskan [13] utilizing a document retrieval framework. With MTE models favoring White-associated names 85.1% of the time and female-associated names 11.1% of the time, they revealed significant biases. Real-world employment prejudice was reflected in the fact that Black males were disadvantaged up to 100% of the time. This study emphasizes the necessity of equitable employment practices for AI and policy issues. By enabling the dynamic retrieval of varied and up-to-date recruiting criteria and reducing reliance on static, historically biased datasets, Lo

et al. [7] also identified bias from unbalanced training data as a major problem and suggested RAG as a solution.

Cai et al. [14] examined how applicants view fairness when AI, rather than people, screens resumes. In their experiments, they found that participants viewed less procedural and distributive fairness when the screener was AI (mean procedural fairness: $M_{human} = 3.92 > MAI = 2.63$; mean distributive fairness: $M_{human} = 3.64 > MAI = 2.49$). This highlights the importance of transparency and trust in AI-driven recruitment processes.

3 Research Methods

This section elaborates on the methodology, addressing dataset preparation, model optimization, summarization, decision-making, and evaluation. The workflow integrates advanced LLMs, hybrid techniques, and human-in-the-loop validation to ensure robustness, fairness, and transparency. The research will automate resume screening by first importing and anonymizing resumes through PII removal. Advanced LLMs classify and filter resume content using multi-stage checks (e.g., eligibility, skill matching). Resumes are then summarized via hybrid methods, and a multi-agent system (Screening, Grading, Decision Agents) evaluates candidates. The system's performance is validated using classification metrics (Accuracy, Precision, Recall, F1, MCC). Candidates are scored via weighted criteria: Experience (40%), Skills (30%), Education (15%), Certifications (10%), and Projects (5%), balancing automation with fairness and transparency.

3.1 Dataset Collection and Preprocessing

3.1.1 Data Collection

The data used in this research was sourced from a public repository on Kaggle that contains a rich and realistic set of resumes from many professional fields. The dataset represents a representative corpus for the development and assessment of intelligent resume processing systems with large language models (LLMs) and rule-based decision systems.

The dataset includes over 2,200 resumes, labeled with a unique job category label, thereby supporting supervised and semi-supervised machine learning experiments. The resumes are collected from real-world sources and not artificially generated and have been anonymized to protect individual identities. The resumes comprise unstructured text data with sections such as Education, Skills, Experience, Certifications, and other qualifications.

A prominent characteristic of this dataset is domain variety. It encompasses 24 professional categories, a broad range of industries such as (Table 1):

1. High-volume technical fields like Information Technology (120 resumes) and Engineering (118)
2. Business-related functions like Business Development (120), Finance (118), and Sales (116)

Table 1. Summary of existing works in automated resume screening

Paper	Task	Dataset	Method	Performance	Findings
Lo et al. [7]	Multi-agent framework for resume screening	Private dataset (105 resumes)	JDA, RA, EA, DMA agents for semantic job-resume matching	Not reported	Limited validation, lacks multimodal integration
Schnabel et al. [8]	Resume-job relevance assessment	TREC Deep Learning Track 2023	Interactive and static ranking pipeline	Cohen's, F1 score improvements	The Dataset has near-duplicates, limits generalization
Licari et al. [9]	Regulatory resume classification using LLMs	EBA Q&As Test Dataset	Multi-step prompting aligned with EU Regulation No. 575/2013	Improved interpretability and accuracy	Prompt dependency, cross-domain generalizability
Wilson and Caliskan [13]	Bias detection in MTE-based screening	500 resumes and 500 job descriptions	Resume audit study using translated documents	MTEs showed significant gender and racial bias	Bias rooted in model training data
Pina et al. [10]	Eye-tracking-based resume suitability prediction	221 recruiters from various industries	Random Forest, Gradient Boosting	RF: 77.5% ± 0.16, GB: 76.4% ± 0.15	Narrow recruiter scope, limited deployment strategy
Kaushik et al. [11]	AI to boost HR productivity	Not specified	Resume analysis with ML model	Not specified	No empirical validation or dataset details
Albassam et al. [14]	Exploring AI in recruitment	Not specified	Conceptual discussion	Not applicable	No experimental results or methods provided
Sharma et al. [12]	Resume classification using TF matrix	Kaggle resume dataset	Elite Bag-of-Words + classifier	62.60% accuracy	Needs better feature representation methods

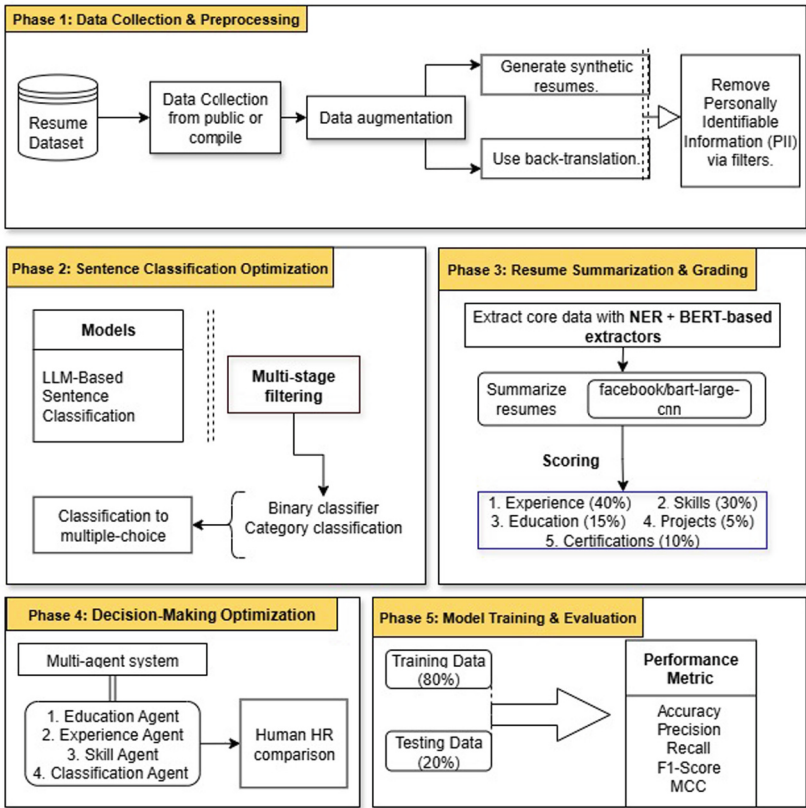


Fig. 1. Proposed framework of resume screening through multi-stage LLM classification

3. Specialized careers like Advocate (118), Healthcare (115), Aviation (117), and Fitness (117)
4. Creative and communication disciplines such as Designer (107), Digital Media (96), and Public Relations (111)
5. As well as less common domains such as Agriculture (63), Automobile (36), and BPO (22)

The methodology followed in this study involves five phases: data preprocessing, sentence classification optimization, resume summary and grading, decision making optimization and model training and evaluation. Figure 1 illustrates the proposed framework.

3.1.2 Preprocessing

Preprocessing plays a critical role in preparing these data for effective classification and summarization. To ensure the ethical use of data, all personally

identifiable information (PII), such as names, contact details, and geographical locations, was removed or obfuscated during the preprocessing step. A structured preprocessing pipeline was executed on the raw resume data to facilitate downstream machine learning and natural language processing tasks, which were initially in HTML format. Every resume was parsed with the BeautifulSoup library and the lxml parser to extract semantically relevant sections. In particular, three fundamental components, like Experience, Skills, and Education, were extracted by finding keyword-based section headers and extracting the following textual content as shown in the Table 2.

Table 2. Three fundamental components after splitting the resume during preprocessing (Experience, Skills, and Education)

	Experience	Skills	Education
0	Respected builder and leader of customer-focused teams; strives to instill a shared	Assistant General Manager Jun 2010–Dec 2010	Business Administration, 1999 Jefferson College
1	09/2015 to current HR Specialist, US HR Operations	Adobe Photoshop, ADP, Asset Management, branding, brochures, content	2014 Master of Arts: Corporate
2	15 plus years in Human Resources Executive Management	Benefit Administration Policy Development, Web Page Development	Bachelor of Business Administration MOUNT VERNON NAZARENE UNIVERSITY
3	HR Generalist/Recruiter January 2013–January 2015	ADP, People fluent, Microsoft Offices, Interviewing expertise Performance management strategies	Masters of Administrative Science: June 2013 Fairleigh Dickinson University

The section wise extraction encourages more targeted representation learning and classification by dividing content that serves distinct semantic purposes. Resumes lacking any of these essential sections were removed to ensure consistency and preserve the integrity of the data. The resultant structured dataset was stored in CSV format, building a neatly structured, labeled corpus suitable for tasks such as section-wise classification, sequence modeling, information extraction, and resume summarization, all critical to automated hiring and advanced career-matching systems [15, 16].

3.2 Summarization

To reduce input complexity and enhance model interpretability, each section of the resume was independently summarized using the facebook/bart-large-cnn model, a variant of the BART architecture fine-tuned for abstractive summarization.

1. **BART:** BART is a transformer model that is highly successful at translation and summarization that has been applied in natural language processing. Its bidirectionality enables the model to be highly competent in reading texts in tandem with contextual sense. Its autoregressive property enables the model to produce coherent and fluent summaries upon being shown the source text. The BART model has been pre-trained using a vast corpus of data and further fine-tuned using task-specific data. It will thus be in a position to generate exceptionally in the field of translation as well as summary [17].
2. **Facebook/bart-large-cnn:** BART-Large is among the versions of the BART (Bidirectional and Auto-Regressive Transformers) model, a strong language-generating model. BART-Large's larger model size enables the model to learn more complex patterns in the language and hence results in better understanding as well as generating language [17]. Facebook/Bart-Large-CNN, after fine-tuning on CNN/DailyMail summarization data set, achieved state-of-the-art performance in terms of standard metrics like ROUGE and BERTScore. It can be transferred for broad-based application for abstractive summarization tasks in any kind of domain.

3.3 LLM-Based Sentence Classification

Bart-large-mnli model [18] that has been fine-tuned by the MultiNLI dataset has, in fact, been utilized in zero-shot classification through the use of the candidate labels as the hypotheses of inference, being a functional, if popular, solution in the new NLP systems posed in [19]. Having been trained upon the tweets, affective events, and folk tales, in addition to the artificially generated sentences, the model has nine emotions (anger, disgust, fear, guilt, joy, in love, sad, ashamed, and surprising) plus the class "none" (none of the above emotions). The solution indicates the sequence to be labeled as the multi-genre natural language inference (MNLI) and creates the hypothesis using all the possible labels. Label probabilities are then built by the probabilities of the entails in addition to the contradiction. It transforms classification problems into text entailment problems: summary abstracts in resumes are made premises, while hypothesis labels such as "Qualified" and "Not Qualified" are made hypotheses. The model provides an estimate of confidence for each label, making inference possible without fine-tuning for a specific task. It provides adaptive, transferable decisions in low-resource environments.

3.4 Decision Optimization via Multi-agent Logic

In a move towards making decisions consistent and accounting for different dimensions when evaluating a candidate, a multi-agent decision system emerged. It integrates the outputs from four different appraisal agents, each with a different dimension in a resume abstract being evaluated:

- The Education Agent assesses educational qualifications in terms of a looking for degree-type terms (e.g., “Bachelor”, “Master”, “PhD”) appearing in the abstracted education field, with binary ranking based on the presence of an advanced-degree.
- The Experience Agent makes an inference of experience from extraction and incrementally adding numeric values that are then assumed to quantify years of experience. Applicants with experience of, or more than, a specified threshold are marked positively.
- Skill diversity Agent approximates skill diversity based on the number of distinct skills in the skills summary section, normalizing breadth to the pre-specified threshold level.
- The Classification Agent uses LLM-based end-to-end semantic processing with the help of a pre-trained zero-shot language model (facebook/bart-large-mnli). It classifies the resume summary as a whole as “Qualified” or “Not Qualified” based on contextual entailment between candidate labels and input.

This overall decision is determined through a four-agent, consensus-based voting system, in which a candidate is designated as “Qualified” when three out of four agents decide with an affirmative conclusion. This four-agent approach combines rule-based heuristics and deep inference in language in a move towards finding a stronger, though more interpretable, decision strategy than an individual agent.

3.5 Model Evaluation and Validation

In the absence of actual HR-labeled resume data to test the ranked pipeline for resumes, a synthetic ground truth is used, simulating realistic HR practice decision-making.

3.5.1 Artificial HR Labeling Technique

Since there were no manual HR labels, we made an implementation with a rule-based labeling function, `HR_Label`, in a simulation of standard HR screening practice. It is extracted from the resultant resume score, providing dominant resume factors, experience, education, and skills. Among them were the following assumptions when designing the HR labeling rule:

- High-confidence qualification ($Score \geq 70$): Candidates with scores above 70 were clearly good. Such applicants were immediately classified as being a “Qualified” (`HR_Label = 1`), corresponding to situations where HR professionals shortlist unambiguously good profiles with no need for any filtering beyond.

- Examples in the borderline range (50–69 score): These mid-level applicants were susceptible to variation in likelihoods. In an effort to replicate the subjectivity common in manual HR shortlisting, a randomized judgment had been made: approximately 70% of these applicants were assigned a mark of being “Qualified” and the rest a mark of being “Not Qualified”. This maintains an in-reality hold insofar as there are profiles shortlisted due to perceived potential, even when potentially unremarkable on paper.
- Low-confidence rejection ($Score < 50$): Scores less than 50 were labelled as a “Not Qualified” category ($HR_Label = 0$) as a standard screening-out choice for résumés with poor qualifications or corresponding experience.

This semi-deterministic approach also provides HR_Label with the potential for being a pseudo-ground truth to measure auto models’ performance.

3.5.2 Comparative Assessment of Classifiers

These previously present category systems were compared to the synthetic HR_Label

1. Rule-Based Classifier:

This classifier only made qualification decisions based on resume score. By a score of ≥ 55 , the candidate qualified; otherwise, they did not. It constituted a baseline for deterministic scoring logic with no contextual comprehension.

2. Multi-Agent Decision System:

This framework engaged four individual appraisers, such as education, experience, diversity in skills, and a big language model (LLM) semantic classifier. Three among the four agents could qualify a candidate with the label “Qualified.” It is a tool like a more thorough and robust evaluating system in which heuristic rules are filled in with the semantic level reasoning.

Each of the classifiers’ performances was evaluated with the following measures: accuracy, recall, precision, F1-score and Matthews Correlation Coefficient (MCC). These metrics provided a general depiction of classification performance under conditions of simulated HR decisions.

3.5.3 Resume Score Predictive Capacity

In order to see if the resume score in isolation truly holds any predictive power, a logistic model was also trained on the feature alone. It was cross-validated with 5-fold stratified cross-validation in an effort to properly determine performance on any arbitrary candidate set. Metrics for accuracy on each fold were investigated in an attempt to determine if a score in isolation could truly replicate HR decision-making patterns, or if there were other advantages being afforded through the more nuanced reasoning with multiple agents.

4 Experimental Results and Discussion

This section contains the description for the quantitative outcome from the comparison between two classifying approaches: the rule-based score model and the multi-agent decision model. The section contains the influence of the distribution of the resume score and the model response with the simulated HR ground truth.

Figure 2 shows the distribution of calculated resume scores in the dataset. That distribution is strongly biased toward the bottom score values, and most candidate profiles had scores in the range below 50. That mirrors the real-world variability in quality in the submitted resumes and accounts for the conservative behavior of both classifiers.

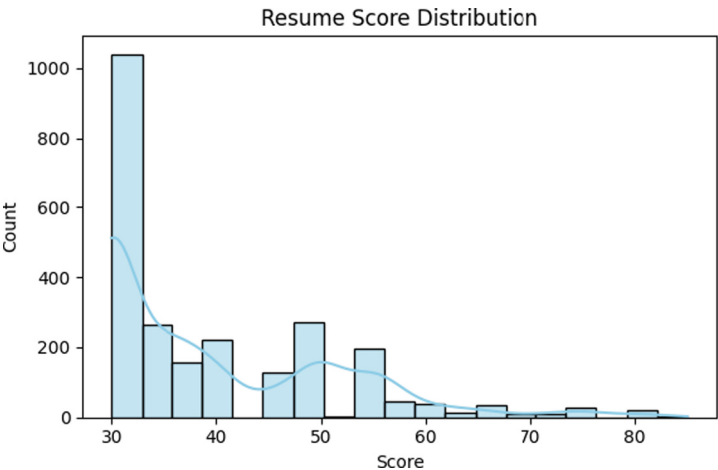


Fig. 2. Histogram showing the distribution of computed resume scores across the dataset

Figure 3 compares the rule-based and the multi-agent classifiers on five performance criteria: Accuracy, Precision, Recall, F1-Score, and Matthews Correlation Coefficient (MCC). The rule-based classifier that classified the candidates as “Qualified” if and only if the resume score was 55 or more performed significantly better with 88.77% accuracy, 78.39% precision, 61.78% recall, 69.10% F1-score, and 0.6302 MCC. The multi-agent classifier even though coded for the exploitation of varied evaluation signals such as education, experience, skills, and LLM-semantic analysis registered lesser values for all the criteria with 82.29% accuracy, 64.98% precision, 27.92% recall, 39.06% F1-score, and 0.3432 MCC. The maximum disparity was in the area of recall, suggesting the multi-agent system would under-identify the potential candidates. This may possibly have been due to the overly conservative model’s consensus mechanism to reach a

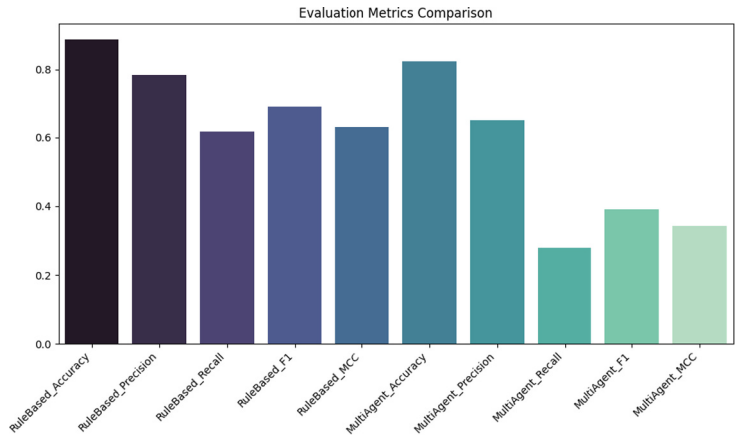


Fig. 3. Evaluation metrics comparison between rule-based and multi-agent classifiers

consensus decision by at least three out of the four agents, which tends to make it too stringent and raise the false negative ratio.

The confusion matrix between the rule-based classifier and the simulated ground truth (HR.Label) shown in Fig. 4. The confusion matrix shows the classifier correctly classifies 312 true positives (Qualified candidates) and 1,893 true negatives (Not Qualified), and incorrectly classifies 86 false positives and 193 false negatives. Such a distribution shows the rule-based model had been extremely effective in discerning the qualified from the unqualified candidates. That many false negative resumes picked by the HR logic as “Qualified” but not by the model points out the strength in the borderline cases in the 50–69 score range. Those cases had been designed with probabilistic uncertainty intentionally in the label, and the misclassification shows the weakness in the deterministic model in dealing with subjective or fuzzy candidate profiles.

Table 3. Comparative performance metrics of classifiers

Classifier	Accuracy	Precision	Recall	F1-score	MCC
Rule-based classifier	88.77%	78.39%	61.78%	69.10%	0.6302
Multi-agent decision system	82.29%	64.98%	27.92%	39.06%	0.3432

A direct comparison of the rule-based and multi-agent classifiers across the five performance criteria is shown in Table 3. The evaluation shows that the rule classifier based on the strength score from the resume has great precision (78.39%) and has great accuracy (88.77%) in providing a strong baseline in filtering out the candidate. With even-balanced recall, the classifier can significantly capture candidate prospects but has low false positive occurrences. Despite the

richer theory in the multi-agent system through the combining the heuristic rules and the semantic reasoning proposed by the LLM, the performance was poorer in composite F1-score and in the recall (27.92%). That is probably the reason why the hard consensus threshold has rendered the system too conservative and generated false negatives. All the same, the modularity and the explainability have the potential in the extreme-stakes or the low-numbers-screening tasks where the precision dominates. While the rule-based approach is more complete for large-scale filtering tasks in general, the multi-agent system requires more flexibility in the actual applications in the real-world HR environment, i.e., more relaxed voting rules or confidence weighting for more flexibility and better recall.

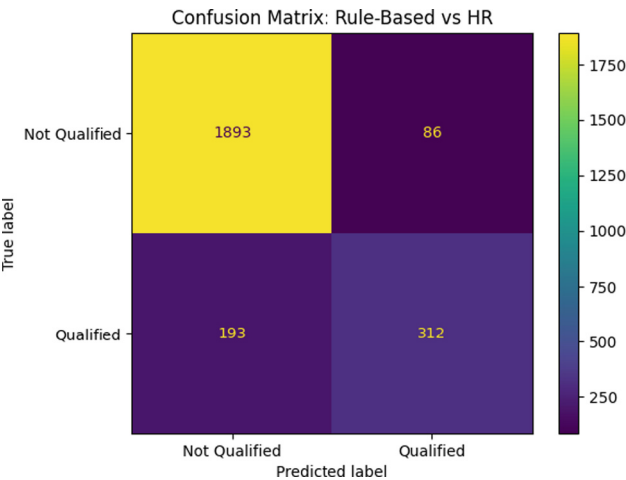


Fig. 4. Confusion matrix of the rule-based model compared to HR.Label

5 Conclusion and Future Work

This paper provides one pipeline for computer-aided rule-based scoring of resumes, LLM-assisted summarization, zero-shot classification, and multi-agent decision-making. The experiments demonstrate the rule-based model provides a good, interpretable baseline with sufficient accuracy, but the multi-agent one still has more semantic reasoning but at the cost of more tuning in order to have the greater recall.

We plan in the follow-up work to increase the system reliability by getting real-world resumes from the actual recruiting companies and actual HR choices as real-world recruiting practices ground-truth annotations. Such will facilitate more reliable validation and training and consistency with real-world recruiting practices towards the creation of more intelligent and scalable screening tools from real-world data.

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